

UDK 621.313.322

**TURBOGENERATORS OF NEW GENERATION WITH VARIOUS
COOLING SYSTEMS**

Minko A.¹, candidate of technical sciences, engineer

Shevchenko V.², candidate of technical sciences, professor

¹Private scientific and technical firm "Ankor-Teploenergo", Kharkov, Ukraine

*²National Technical University "Kharkiv Polytechnic Institute",
Kharkov, Ukraine*

At present, power equipment that has fulfilled its technical resource requires not carrying out repair and restoration works, but full replacement with new, high-tech and competitive equipment. There is a need to create and manufacture generating units that meet modern requirements and trends in the development of the electric power industry, [5]. The basic requirements for new turbogenerators are, [4]:

- Increase of generators electric power with preservation of former dimensions;
- Reduction of specific mass, both the generator itself and individual elements of its design;
- increase the generator service life and the time between repairs;
- Increased reliability of the turbo-unit operation and its auxiliary systems (gas, oil, hydraulic seals)
- Reduction in the cost of generating equipment and there structural elements that wear out quickly (seals, oil separators, bearing shells, heat exchangers, etc.);
- Reduction in the cost of generating equipment and the cost of structural elements those wear out quickly (seals, oil separators, bearing shells, heat exchangers, etc.).
- Replacement of hydrogen with air, as a cooling medium, in those cases where it is permissible.

Our theoretical studies and practical experience in the design of turbogenerators, as well as analysis of current literature [1-3, 6], made it possible to review the technical parameters of 250 MW turbogenerators with various cooling media (air, hydrogen, water-hydrogen). This can be used to create turbogenerators of new generation. The data are presented in Table 1. For turbogenerators that will be designed in the future, the choice of the cooling system

type, taking into account the possibility of using different heat carriers, should be carried out using the recommendations set forth in [4]. The indicators listed in the table can be used in the development and manufacture of competitive domestic turbogenerators. Such turbogenerators must comply with international standards for the technologies used, be economical and have high operational reliability.

Table 1.

Technical parameters of 250 MW turbogenerators with different cooling systems, stator voltage 15,75 kV, $\cos\varphi = 0,85$, number of pole pairs $p=1$

Parameters	Indicators		
Cooling Medium			
– stator	air	hydrogen	hydrogen
– stator windings	air	hydrogen	water
– rotor	air	hydrogen	hydrogen
Stator winding current, kA	10,78	10,78	10,78
Single-sided air gap, mm	50	100	77,5
Current density in the stator winding, A/mm ²	2,6	3,9	8,5
Number of stator slots	72	60	30
Stator length, mm	4970	5000	4020
Outer diameter of the stator core, mm	2800	2515	2530
Geometry of the groove in the stator core, mm			
– slot width	27	38,6	50,8
– depth of groove	261	250	183
Active rotor shaft length, mm	5000	5100	4100
Diameter of the rotor, mm	1200	1075	1120
Number of coiled grooves	28	36	36
Geometry of the groove in the rotor, mm			
– slot width	stepped 51×48×44	30,6	33,9
– depth of groove	220	173,5	165
Excess cooling gas pressure, MPa	0	0,4	0,4
The total mass of the turbogenerator, t	210	265	250

Literature

1. Minko A.N. Comprehensive approach to the design of cooling systems for large electric machines (turbogenerators) – Electrical equipment: operation and repair.–2017–№ 1–P.p. 12–18.
2. Minko A.N. Graphic model of the design of an effective cooling system for large electric machines (turbogenerators) / Electrical power through of the eyes youth: materials of the VII International Scientific and Technical Conference, September 19-23, 2016, Kazan. – In 3 tons of T1. – Kazan: Kazan. state power. Univ. - 2016. – P.p. 66–69.
3. Minko A.N. Optimal geometry and mass-dimensional parameters of the stator casing structure of a turbogenerator with an air cooling system / Energy Saving, Energy, Energy Audit. – 2012. – № 1. – P.p. 33–39.

4. Shevchenko V.V., Minko A.N. Development of cooling systems and optimization of turbogenerator designs: monograph. – Kharkov: Publ. Ivanchenko I.S. – 2013. –. 242 p.
5. Shevchenko V.V., Minko A.N. On increasing the competitiveness of domestic turbogenerators // Kremenchug: KrNU. - Collection of proceedings of the XI International Scientific and Technical Conference, Kremenchug, April 09-11, 2013. - P.p. 386–387.
6. Shevchenko V.V., Minko A.N. Optimization of the mass-size dimensions of gas coolers of turbogenerators during the reconstruction and technical re-equipment of power plants // Energetic. – 2016. – № 2. – P.p. 52–54.